

COMPARISON OF RESULTS

Method	Thread s/GPU	Classifier Training (s)	Classificat ion Time (s)	Speed up	Efficiency	Test Accur acy	Balanced Accuracy	Macro F1	Weighted F1
Sequential CPU	1	8.313	8.388	1.00×	100.0%	0.8053	0.6773	0.6235	0.8301
OpenMP CPU	2	5.710	5.733	1.46×	73.15%	0.8053	0.6774	0.6233	0.8235
OpenMP CPU	4	4.101	4.119	2.04×	50.91%	0.8047	0.6769	0.6229	0.8227
OpenMP CPU	8	3.007	3.026	2.77×	34.65%	0.8053	0.6774	0.6233	0.8234
OpenMP CPU	12	3.289	3.310	2.53×	21.12%	0.8053	0.6773	0.6235	0.8235
CUDA (Tesla T4)	GPU	0.392	—	—	—	0.8293	0.6832	0.6661	0.8301

Feature extraction:

Platform	ESM Extraction (s)	Structural Extraction (s)	Total Pipeline (s)
CPU	1094.67	280.82	—
CUDA	78.766	439.152	518.31

Sequential CPU, OpenMP :

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FINAL CPU + OPENMP RESULTS											
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mode	threads	training_time_seconds	total_time_seconds	speedup	efficiency	epochs	validation_accuracy	test_accuracy	test_balanced_accuracy	test_macro_f1	
Sequential CPU	1	8.313459	8.387528	1.000000	1.000000	40	0.803333	0.805333	0.677274	0.623461	
OpenMP CPU (2 threads)	2	5.710156	5.733133	1.462992	0.731496	40	0.802667	0.805333	0.677375	0.623347	
OpenMP CPU (4 threads)	4	4.100543	4.118877	2.036363	0.509091	40	0.803333	0.804667	0.676910	0.622926	
OpenMP CPU (8 threads)	8	3.007389	3.026055	2.771769	0.346471	40	0.803333	0.805333	0.677375	0.623347	
OpenMP CPU (12 threads)	12	3.288708	3.310013	2.533986	0.211166	40	0.802667	0.805333	0.677274	0.623461	

CUDA (Tesla T4 on google colab):

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FINAL CUDA RESULTS
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GPU: Tesla T4
ESM features: 320
Structural features: 28
Combined features: 348
ESM extraction: 78.766 seconds
Structural extraction: 439.152 seconds
Classifier training: 0.392 seconds
Total runtime: 518.31 seconds
Validation accuracy: 0.826
Test accuracy: 0.8293
Balanced accuracy: 0.6832
Macro F1: 0.6661
Weighted F1: 0.8301
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